

how does the thought of leasing the remaining 90 strike you? While that's not exactly accurate, Microsoft Research's Desney Tan and Pradeep Shenoy from the University of Washington do plan on using your mind to help computers detect and recognise faces better. Face detection algorithms are extremely complex, and depend on camera quality, the angle that the face is photographed at, the background and what-have-you. A picture with five faces in it may well get rejected as junk. We, on the other hand, can pick a face out of a crowd, at night, *and* with our eyes half-closed. Incidentally, the human brain isolates faces subconsciously—you could be sitting in a crowded bus, reading the newspaper, but your brain is processing all the faces that enter your peripheral vision.

And that's where *Subconscious Computing* comes in. Tan and Shenoy's system records subjects' EEG (Electro-encephalograph, or brain-waves) patterns while they are viewing images—the responses are starkly different when they view images with and without human faces. With a single subject, the computer was able to identify images with faces with 72.5 per cent accuracy; for eight subjects, it's 98 per cent. Imagine this system hooked up to a security guard watching a surveillance video—using his EEG waveform, the system can determine

The idea is based on the fact that a single human step can power two 60-watt light-bulbs for a whole second. What if you multiplied this to tens of thousands of people, taking tens of thousands of steps so many times a day? Graham and Jusczyk have the idea of a railway station with movable tiles on the floor—people walk, the tiles get depressed and power a dynamo, which generates electricity. "Crowd Farming," as they call it, will use the movement of crowds—especially at rush hour—to generate power, which can then be used for lighting signboards, they say. Get them a large enough crowd (and we have so many of those in India, don't we?), and maybe we can power whole sections of the railway station! The idea also extends to rock concerts—as fans jump in frenzy, they could be powering the artists' amplifiers.

Their working prototype is a stool in a train station in Torino, Italy; people sit on it, causing four LEDs to light up. Silly, but it's the idea that counts. The trouble is that when the crowds stop, so will the power; even assuming that the energy is being stored in batteries, there's only so much they can store with today's technology.

Speaking of batteries, it might be that we won't need them anymore—not for our personal gadgets, anyway.

The Machines

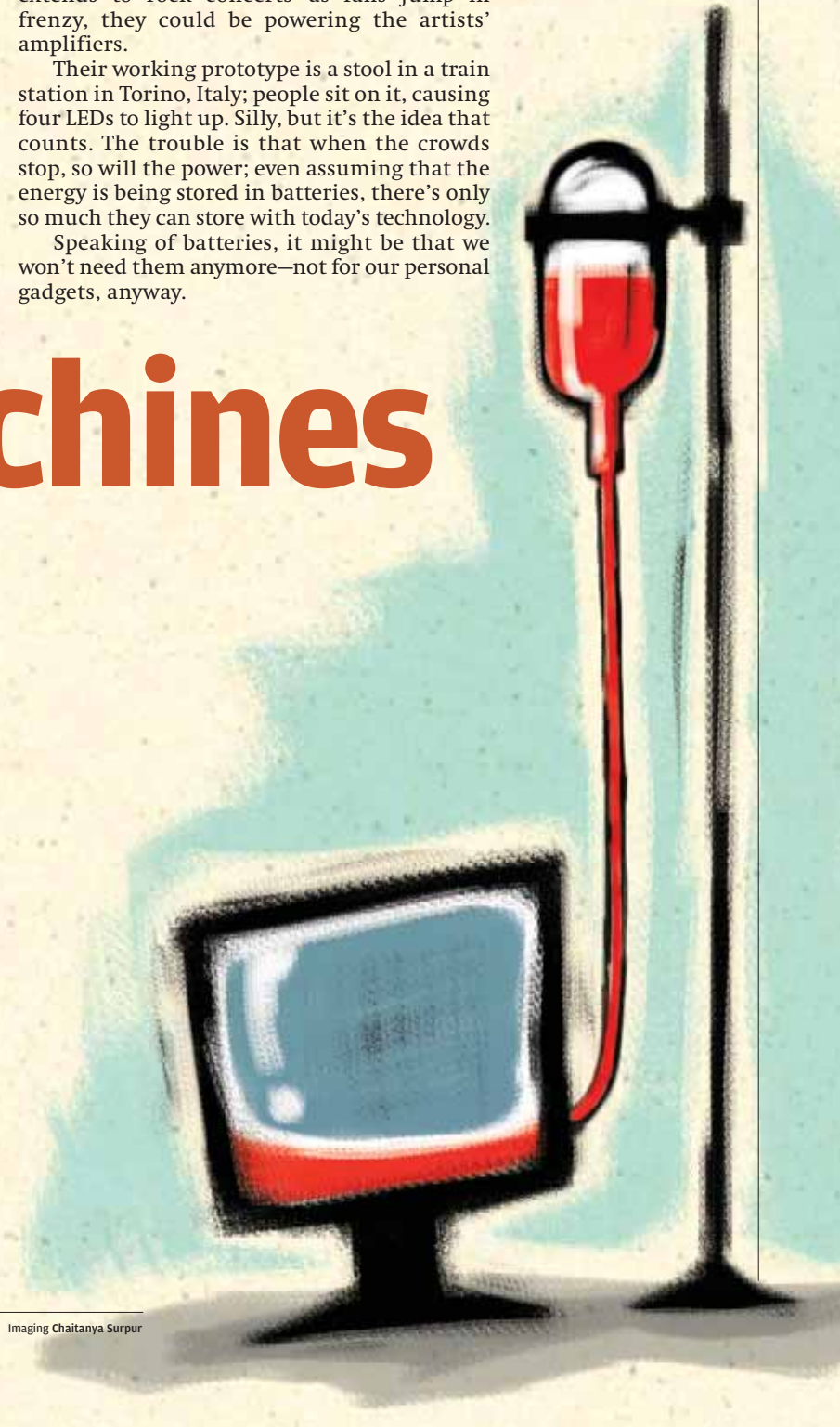
which frames have faces in them, so if the feed needs to be analysed, hours of useless footage can be safely discarded.

Sure, it's spooky—we don't want machines to have access to our subconscious minds (hell, we barely know what's going on in there), but the concept is in the nascent stage, so we've got plenty of time to sort out the ethical issues: one of the ideas is to have images flash on screens in your peripheral vision while you go about your daily business—so you aren't interrupted, and the system gets the output it wants. Of course, this can get pretty intrusive.

Not everything needs to interfere with Life As Usual...

People Power

You've probably seen bicycle headlamps powered by the wheels' rotation, which in turn are powered by the cyclist's pedalling. The OLPC Project's XO laptop is powered by a hand-cranked battery charger. Conservation of energy—it goes from one body to another, from one form to another, and makes all this possible. When we walk, we release energy into the ground; nothing happens. Well, nothing immediately obvious, anyway. James Graham and Thaddeus Jusczyk, students at MIT's School of Architecture and Planning, plan to make something happen.



Imaging Chaitanya Surpur